

Very low noise transimpedance amplifiers to readout SiPMs at cryogenic temperature

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(With many thanks to the DUNE PDS and LHCb RICH collaborations)

Workshop on Electronics for physics experiments and applications @INFN

5 – 7 Mar 2025, Torino, Italy

SiPMs and transimpedance amplifiers

SiPMs: solid-state detectors, sensitive to single photons

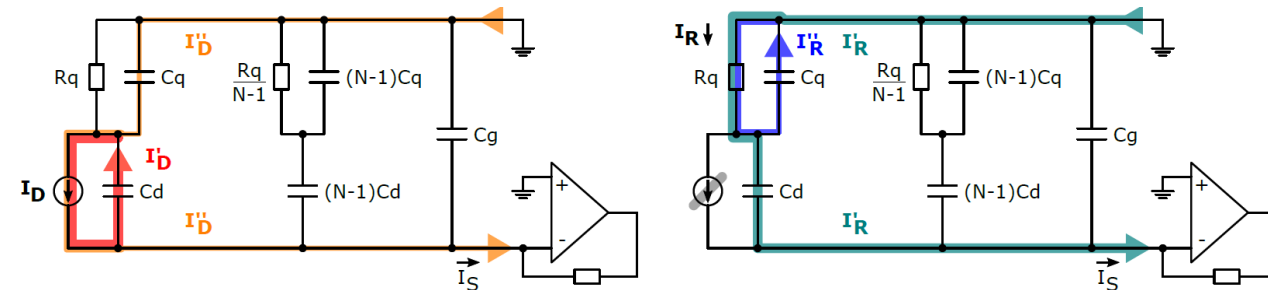
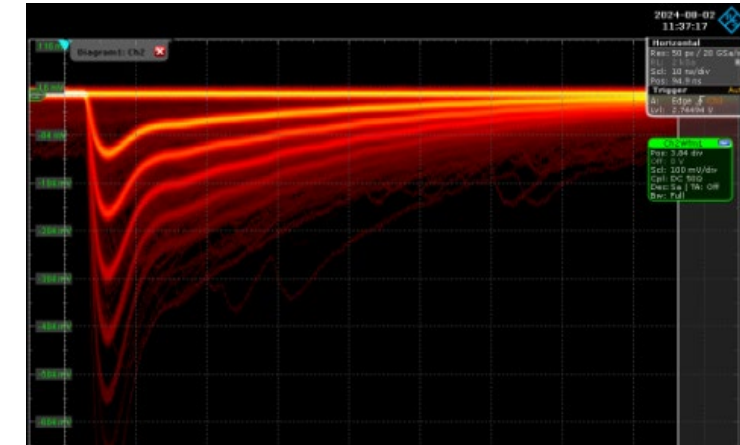
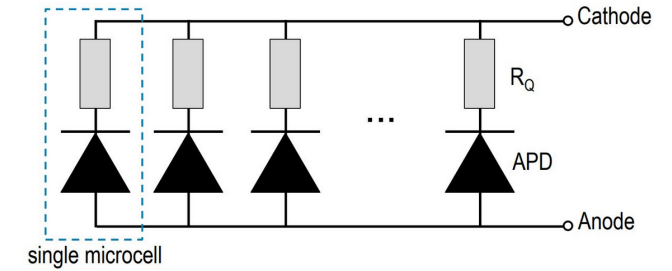
- Single cell: a SPAD in series with a quenching resistor ($C_d \approx 200$ fF, $R_q \approx M\Omega$)
- Thousands of cells in parallel: ≈ 50 pF/mm² + k Ω *mm²
→ Even a small SiPM presents a relatively **low source impedance**

Signals have a **fast rise time** and a **slower recharge time constant** ($\tau = C_d R_q$)

- Two components may be visible, a fast delta-like spike (due to C_q) followed by an exponential tail
- A resistance in series with the SiPM makes the signal last longer and reduces the amplitude, which is typically undesirable → Need an amplifier with a low input impedance
- Given the signal shape, current integration (i.e. using a charge-sensitive amplifier) is often not the preferred choice, compared with direct trans-R conversion from current to voltage

Readout with a **transimpedance amplifier**:

- Converts the SiPM current to a voltage $V_o(t) = -R I_{in}(t)$
- Read the current on a low impedance node
→ SiPM terminals at fixed V
- Preserves the shape of the current signal
- Many SiPMs can be connected in series and/or parallel without affecting the signal shape, fall time, ...



SiPM model: <https://doi.org/10.1016/j.nima.2006.10.219>

Low noise at cryogenic temperature

Different applications, similar requirements of the front-end amplifier:

DUNE Photon Detection System

- 20 kton LAr time projection chamber, operation in liquid Argon (87 K)
- Resolution to single photoelectrons, rise time below 100 ns → Low voltage noise
- Each channel can range up to 2000 mm² of SiPM area; SiPMs in parallel and/or in series («ganging»)
- Source impedance of $\approx 10\text{-}1000\text{ nF} + 100\text{-}1\text{ }\Omega$ at the input of the amplifier
- Low input impedance → Low feedback factor → Need high gain-bandwidth product

LHCb RICH Upgrade 2

- Single photon counting with pixels of 1x1 -3x3 mm²
- Neutron fluence $\approx 10^{13}\text{ n}_{\text{eq}}/\text{cm}^2$ → SiPMs cooled to low temperature (down to 77 K)
- Time resolution better than $\approx 100\text{ ps}$ (sigma) → Wide bandwidth, low noise
- An amplifier is needed for detailed SiPM characterization between 77-300 K (R&D effort) without waiting for the ASIC that is being designed for RICH Upgrade 2

Amplifier requirements:

- **Low voltage noise e_n**
- **High gain-bandwidth**
- **Operation at cryo T**

- **Low voltage noise e_n**
- **High gain-bandwidth**
- **Operation at cryo T**

Commercial opamps can be a good option, but:

- Power consumption of low noise, wide bandwidth opamps is in the 100 mW range (10-15 mA/amplifier)
- Not much flexibility in tuning the gain-bandwidth product, for optimal bandwidth and feedback stability

→ Next best (easiest) thing: **Transistor + opamp**

- A well-known topology that is particularly suited for this kind of applications
- Bipolar transistor for low **voltage noise e_n** ; current noise is anyway negligible
- Opamp can be chosen for optimal bandwidth and power consumption, irrespective of noise

BFP640 SiGe HBT

Silicon Germanium heterojunction bipolar transistor:

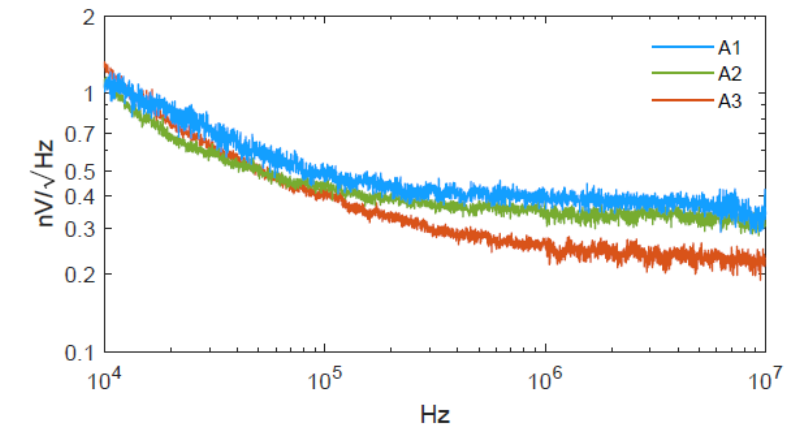
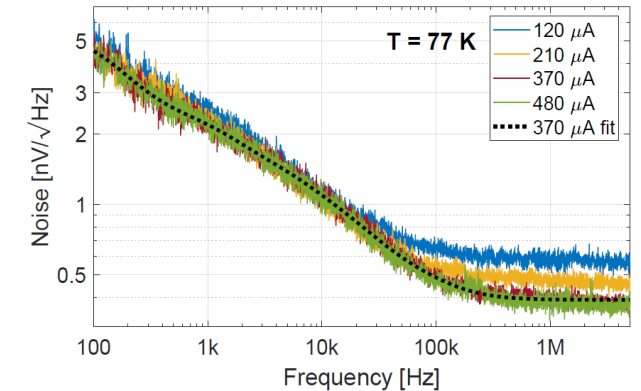
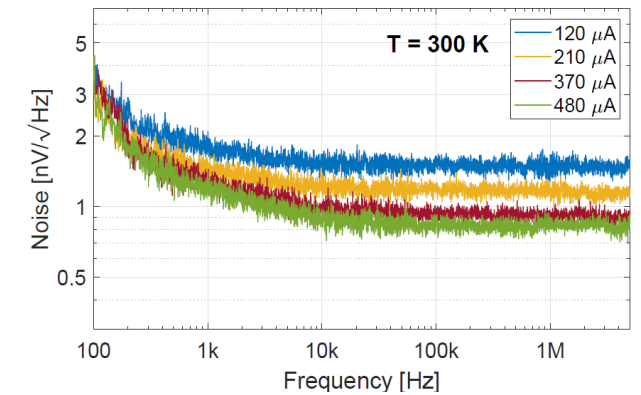
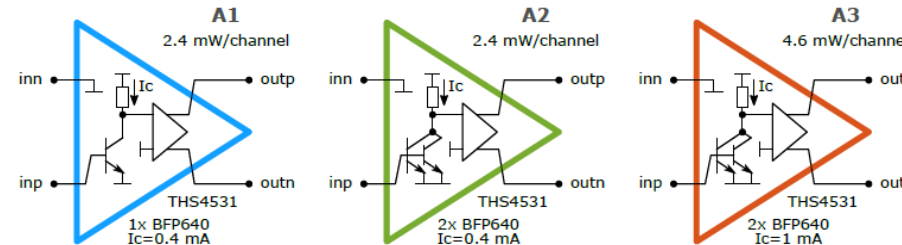
- Designed for radiofrequency $\rightarrow$ low $R_{bb'}$ $\rightarrow$ low thermal noise
- **Wide bandwidth** (a few GHz) even at low current, below 1 mA
- Highly doped base ensures it still functions at cryogenic temperature

Operation at cold:

- Increased g_m $\rightarrow$ lower white noise above 100 kHz
- Essentially unaffected by aging (hot carrier effects, electromigration)
- Higher f_T $\rightarrow$ Need extra care to prevent GHz oscillations:
 - Minimize inductance between the emitter pins and ground
 - Add a small (few pF) capacitor between base and emitter, close to the pins

White noise **0.37 nV/ $\sqrt{\text{Hz}}$** at 77 K:

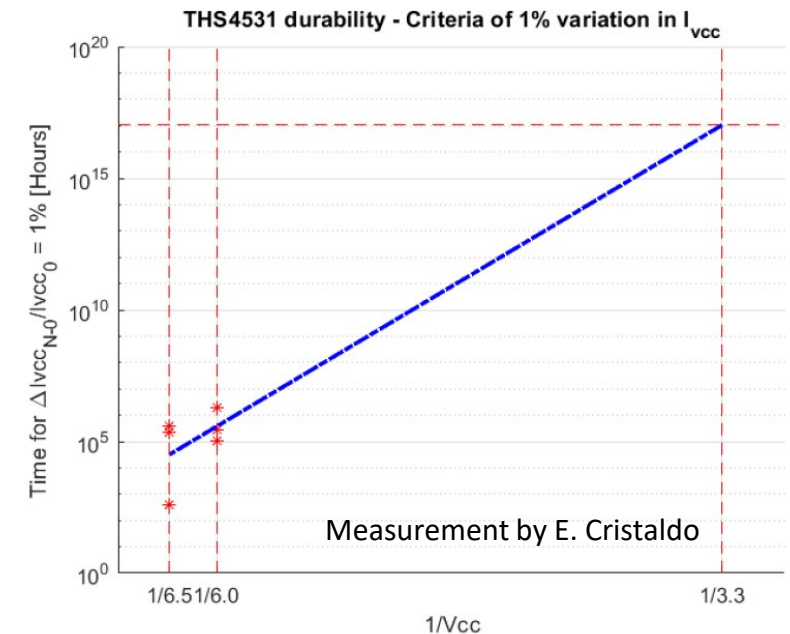
- Does not depend on IC above $I_C \approx 0.4$ mA $\rightarrow$ Effect of $R_{bb'}$, $\approx 20 \Omega$
- Lower noise, down to **0.25 nV/ $\sqrt{\text{Hz}}$** , can be obtained by using two transistors in parallel



CMOS / Bi-CMOS opamps

Commercial CMOS / Bi-CMOS opamps:

- While CMOS transistors are generally functional at cryogenic temperature, a full opamp circuit may or may not work (due to shifts in V_{th} , biasing issues, stability, ...)
- Each opamp needs to be tested
 - Even if a given model works, a wider spread in characteristics may occur between samples at cryo T
- Examples, from colleagues at INFN:
 - AD8651 used in GERDA (after a JFET in a transistor+opamp configuration) → [10.1109/NSSMIC.2010.5873998](https://arxiv.org/abs/10.1109/NSSMIC.2010.5873998)
 - LMH6629 used in DarkSide (SiGe Bi-CMOS opamp, noise comparable with the BFP640) → [10.1109/TNS.2018.2799325](https://arxiv.org/abs/10.1109/TNS.2018.2799325)
 - THS4531 fully differential opamp (see later)
 - LMH6702 current-feedback opamp (see later)
- Lifetime may be impacted by **hot carrier effects (HCE)**:
 - Higher electron mobility at cryo T → higher damage to transistors by impact ionization, compared to room T
 - Expect scaling with V_{DS} as **$\log(\text{lifetime}) \propto 1/V_{DS}$**
 - Mitigation: operating at lower V_{DS} (lower power supply voltage)
 - A decrease of 10% in V_{DS} from the maximum rated supply voltage is usually enough to counterbalance the increased aging at cryo T



DUNE Photon Detection System

DUNE and ARAPUCA photon detectors

The DUNE Experiment:

- High intensity neutrino beam from Fermilab to SURF, under construction
- Modular TPCs, four far detectors > 40 kt fiducial mass
- International collaboration of 200+ institutions

Photon Detection System:

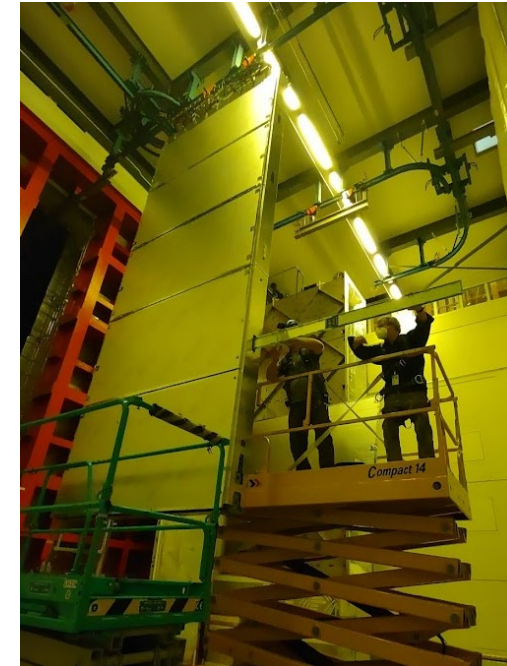
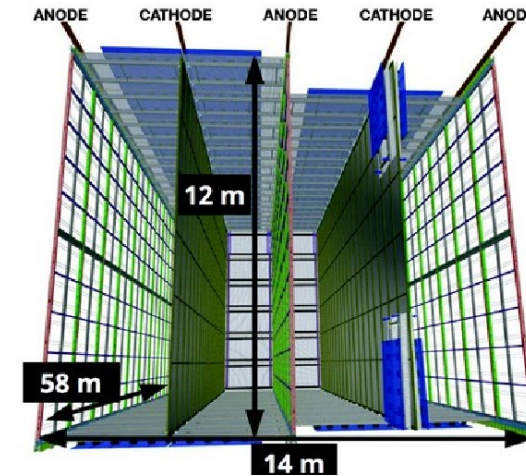
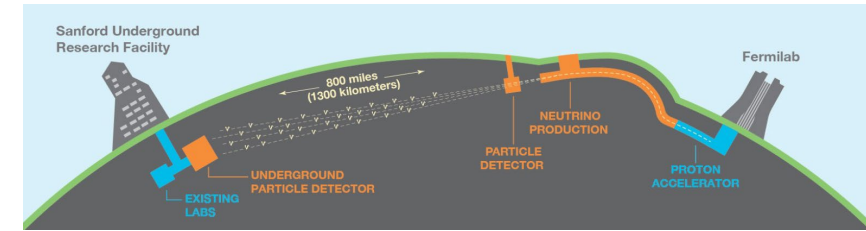
- **ARAPUCA** technology to trap photons on a $\approx 500 \text{ cm}^2$ surface and guide them to 48 $6 \times 6 \text{ mm}^2$ SiPMs ($\approx 20 \text{ cm}^2$) (numbers for FD1)
- For FD1, photon detectors behind the anode wires
- A similar system for FD2, but on the cathode and cryostat walls

FD1:

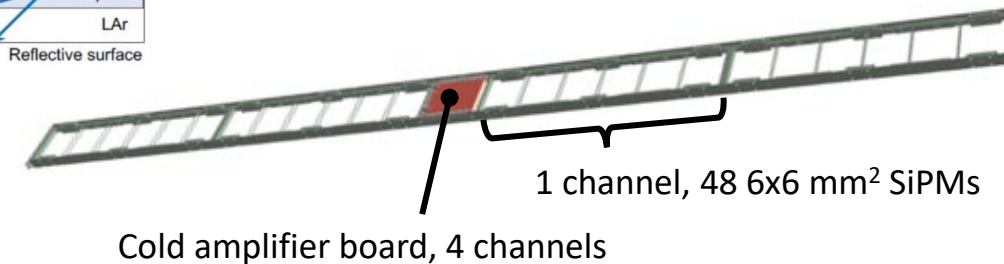
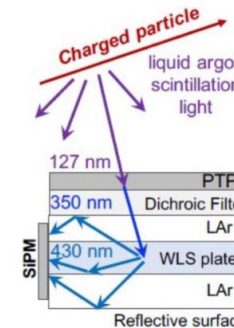
- Each channel is composed of 48 $6 \times 6 \text{ mm}^2$ SiPMs in parallel
- The source impedance is $\approx 60\text{-}90 \text{ nF}$ in series with $\approx 5\text{-}10 \Omega$

FD2:

- Each channel is composed of 80 $6 \times 6 \text{ mm}^2$ SiPMs in «hybrid ganging» (AC-series / parallel combination)
- The source impedance is $4\text{-}6 \text{ nF}$ in series with $\approx 100 \Omega$



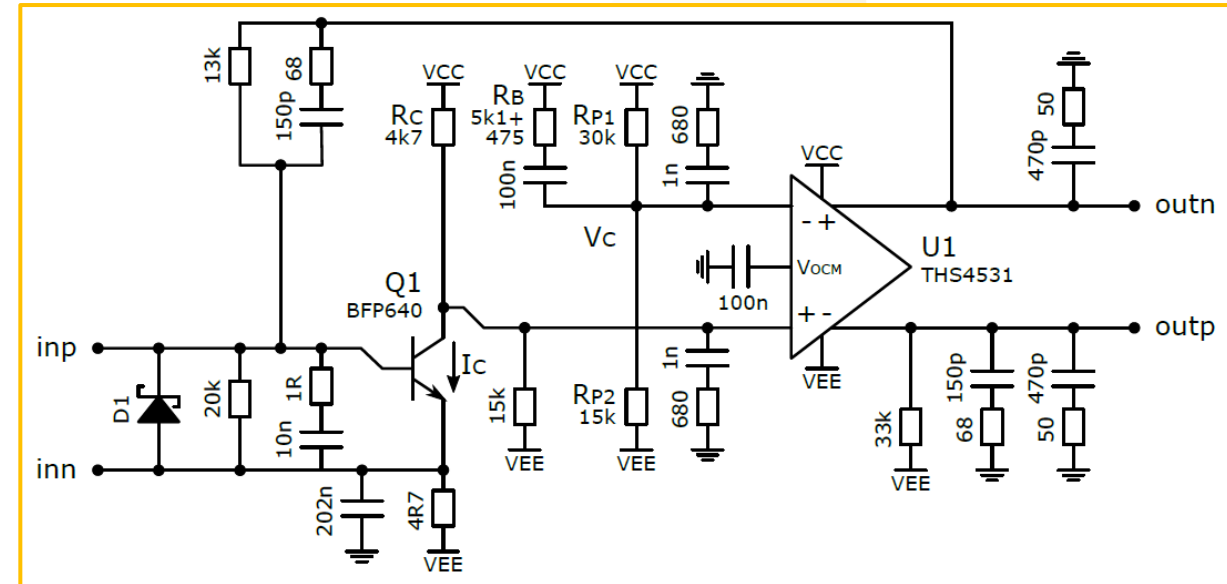
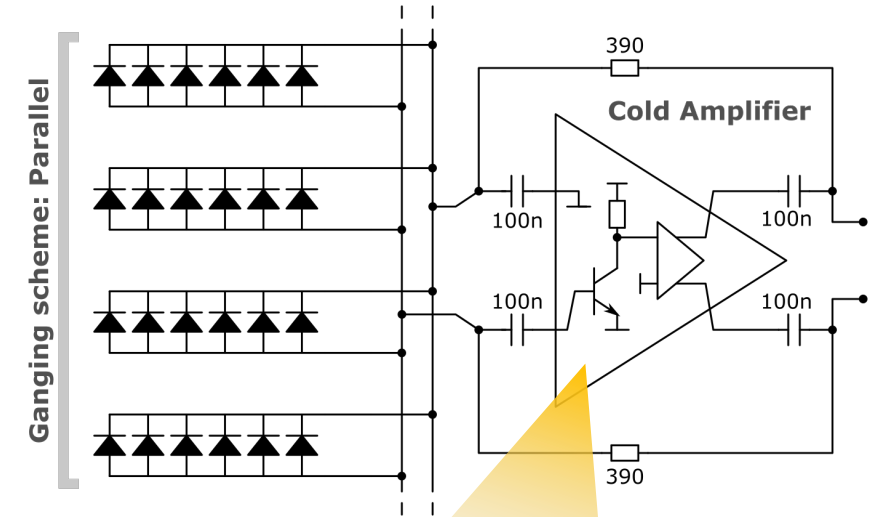
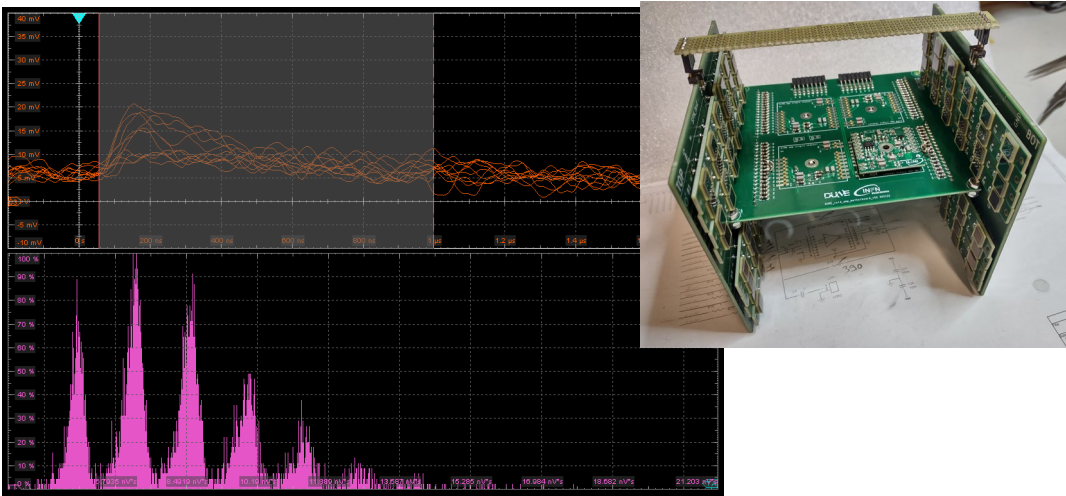
DUNE FD1-HD PSD module



Cold amplifier board, 4 channels

Amplifier design

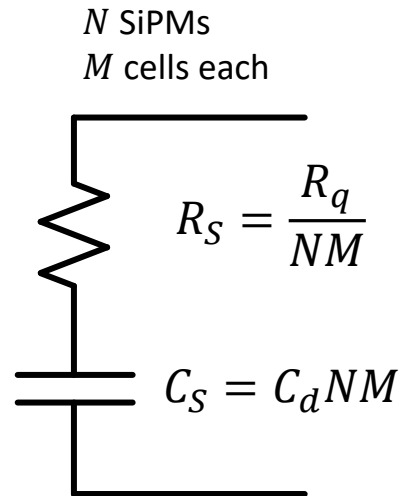
- **BFP640** at 0.4 mA bias current
 - 0.37 nV/√Hz white noise at 77 K
- **THS4531** fully differential opamp
 - Draws 0.25 mA (0.18 mA at 77 K) from a single +3.3 V supply
 - Operated at 60% of the maximum rated supply voltage (+5.5 V) to prevent aging
 - Stress-tested and qualified against hot carrier effects
 - Drives 50-ohm lines (30 m long cables to the DAQ outside the cryostat)
- Closed-loop bandwidth: **5 MHz** (70 ns 10% to 90% rise time)
- Power consumption: **2.4 mW/channel**
- Observed a good single photoelectron resolution with 48 SiPMs in the lab
- Confirmed in the recent ProtoDUNE-2 (HD) run at CERN with 160 channels successfully operated for several months



SiPMs in Parallel

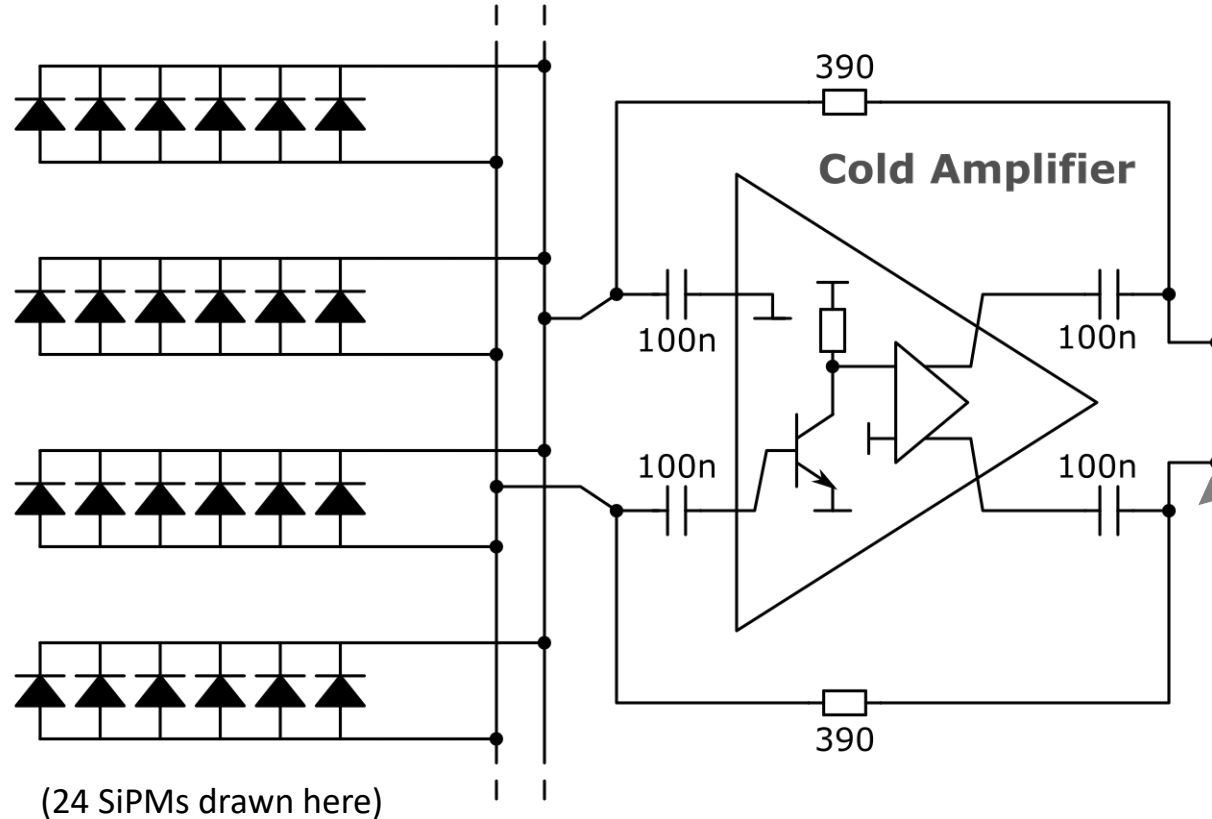
FD1 uses **all SiPMs in parallel**:

- Simplest arrangement (no passive components required around the SiPMs)
- Very low source impedance:
 - Need very low voltage noise
 - Need high open loop gain to keep the input impedance low



$\approx$

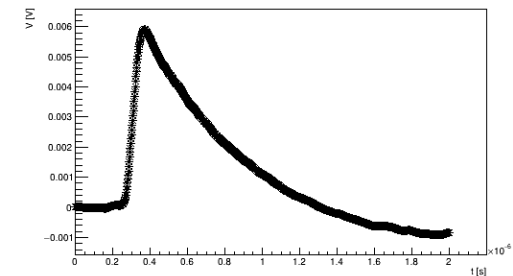
Ganging scheme: Parallel



Current signal from the SiPM
(exponential shape):

$$I_S(t) \approx \frac{V_{OV}}{R_q} e^{-t/\tau_S}$$

$$V_o(t) \approx 2R_f I_S(t)$$

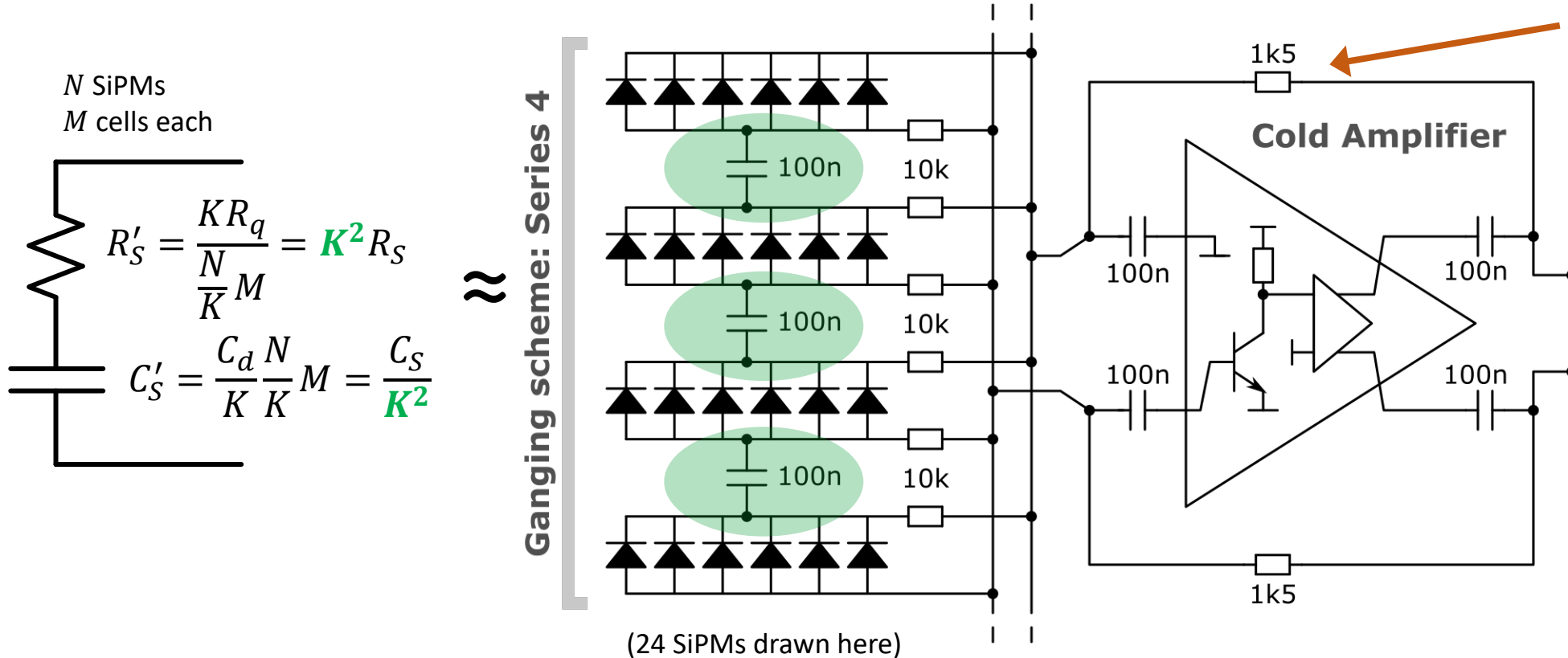


One differential pair used for SiPM
bias (DC) and amplified signals (AC)
→ [10.1088/1748-0221/17/11/P11017](https://doi.org/10.1088/1748-0221/17/11/P11017)

SiPMs in «Hybrid Ganging»

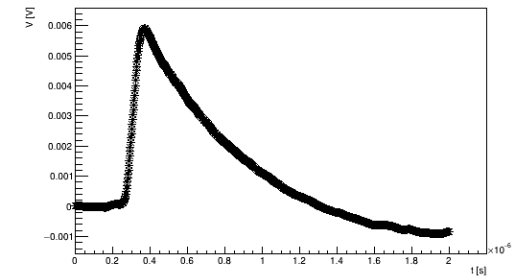
FD2 uses SiPMs in **Hybrid Ganging** (combination of parallel and AC-series connection): → Used in MEG II [arXiv:1801.04688](https://arxiv.org/abs/1801.04688) and in DUNE FD2

- K sub-groups connected in series in AC through capacitors (reduces to parallel ganging if K = 1)
- Signal attenuated by a factor K → Need higher amplifier gain to compensate
- **Larger source impedance** by a factor K^2 for the same total number of SiPMs
→ Reduces the relative weight of series noise and **improves S/N**
- An amplifier with low e_n is still required



R_F is K times larger
to compensate the gain loss

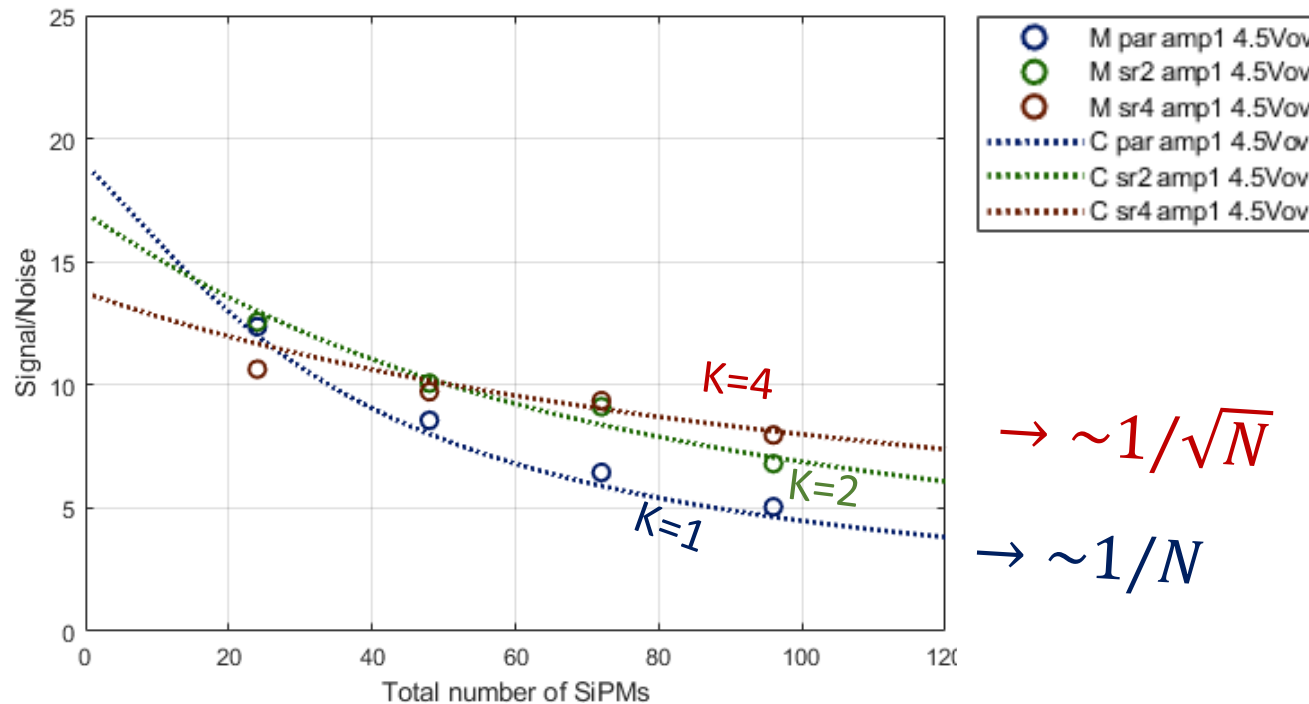
$$V_o(t) \approx 2(KR_F) \frac{I_S(t)}{K}$$



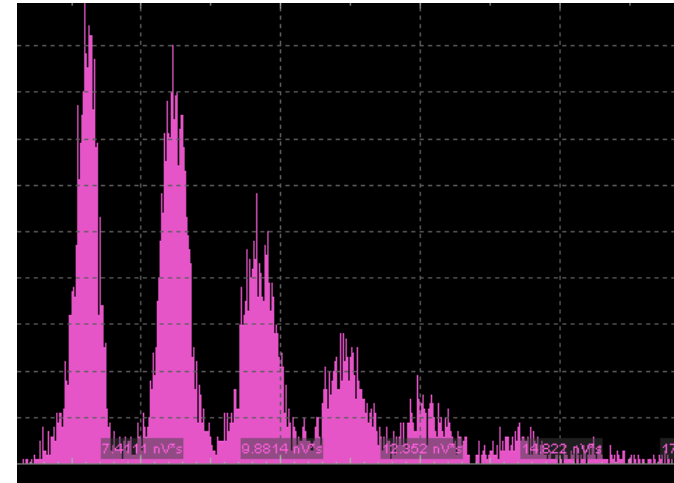
Same signal shape and gain
at the output

SiPM ganging comparison

- **Parallel ganging:** noise scales as the total number of SiPMs
 - S/N inversely proportional to the total input capacitance
 - The voltage noise of the amplifier is the main contributor
- **Hybrid ganging:** noise scales as the *square root* of the total number of SiPMs
 - S/N proportional to the square root of the total input *resistance*
 - The thermal noise of the quenching resistors is the main contributor (if the parameter «K» is high enough; an amplifier with low noise helps in achieving it sooner)



96 6x6 mm² FBK SiPMs @ 4.5Vov
Series4 ganging + Amp1
Signal/Noise ≈ 8



LHCb RICH Upgrade 2

LHCb RICH Upgrade 2 R&D

The LHCb Experiment

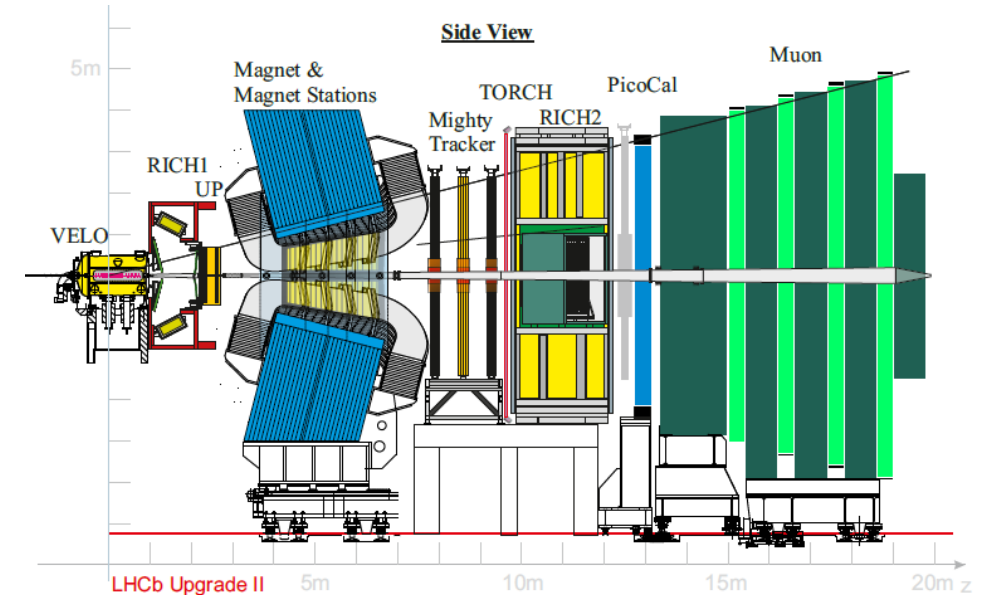
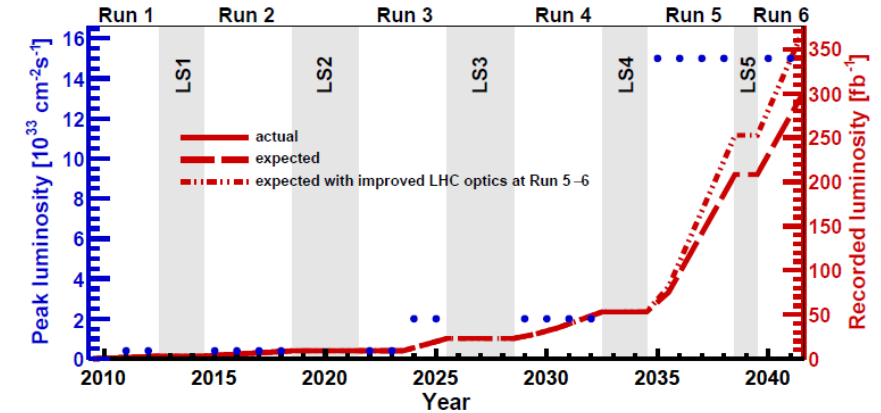
- Upgrade 2 proposed for Run 5 & Run 6
- Significant increase in luminosity
- Higher occupancy in all subdetectors

Requirements of the **RICH (Ring Imaging Cherenkov)** subdetectors:

- **Smaller pixels:** down to $1.4 \times 1.4 \text{ mm}^2$
- **Time resolution:** better than 100 ps RMS
- **Radiation tolerance:** up to a few $10^{13} \text{ n}_{\text{eq}}/\text{cm}^2$
- The RICH Upgrade 2 readout chain will be based on an ASIC (FastRICH) which is being developed in parallel with the choice of photon sensor

Leading photon sensor candidate: **SiPMs cooled to cryogenic temperature**

- R&D is required to choose the best device for the RICH Upgrade 2
- Irradiation and testing campaigns are starting
- Custom runs planned with SiPM vendors
- The R&D campaign needs a dedicated amplifier with:
 - Wide bandwidth
 - Low noise and excellent timing performance
 - Able to operate in the entire temperature range 77-300 K



Noise and timing performance

- Timing performance of the amplifier depends on bandwidth (rise time) and noise:

$$\sigma_t \approx \frac{t_{rise}}{S/N} \propto \frac{1}{Q} \sqrt{\frac{e_n^2}{f_{BW}}}$$

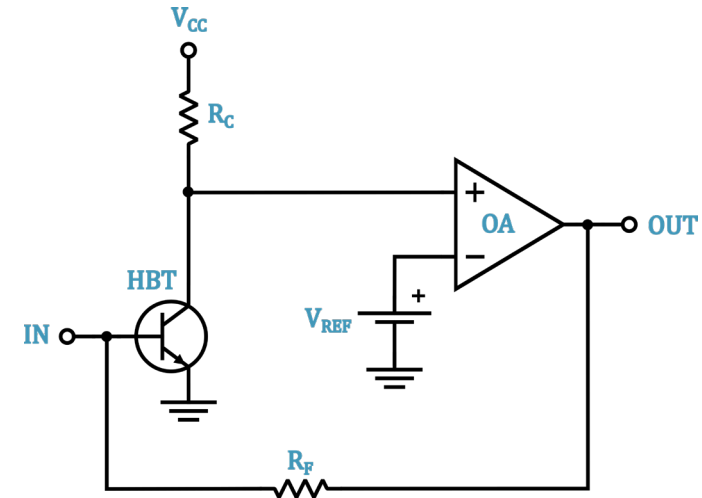
t_{rise} : signal rise time

S/N : signal to noise ratio

e_n : white noise of the amplifier

f_{BW} : closed-loop bandwidth of the amplifier

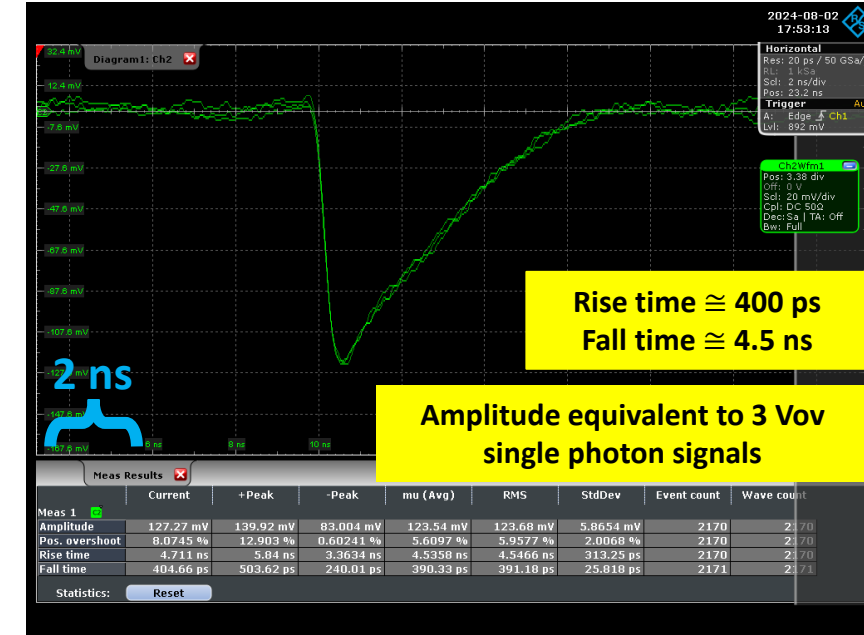
Q : signal amplitude at the amplifier input



- Ideally, this contribution should be made negligible compared with the time resolution of the photon detector

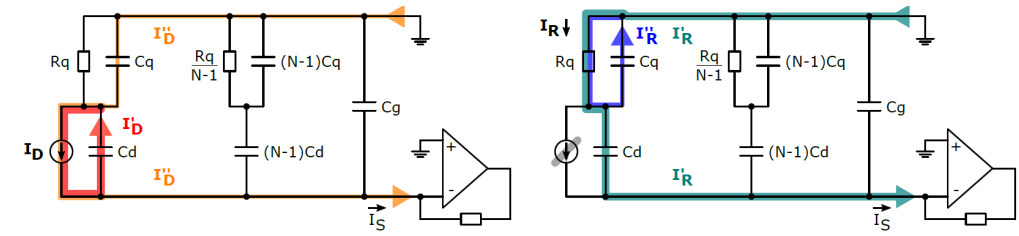
- The BFP640 HBT fits perfectly (low noise, wide bandwidth)
- The operational amplifier needs to be faster than the THS4531 used in DUNE
- A good choice is the LMH6702 (current feedback, 1.7 GHz unity-gain bandwidth) which works down to 77 K

- The collector voltage V_{CC} can be varied to tune the transistor current
→ tune the loop gain → stability and max bandwidth at all temperatures
- ≈1 GHz closed loop bandwidth (400 ps rise time)

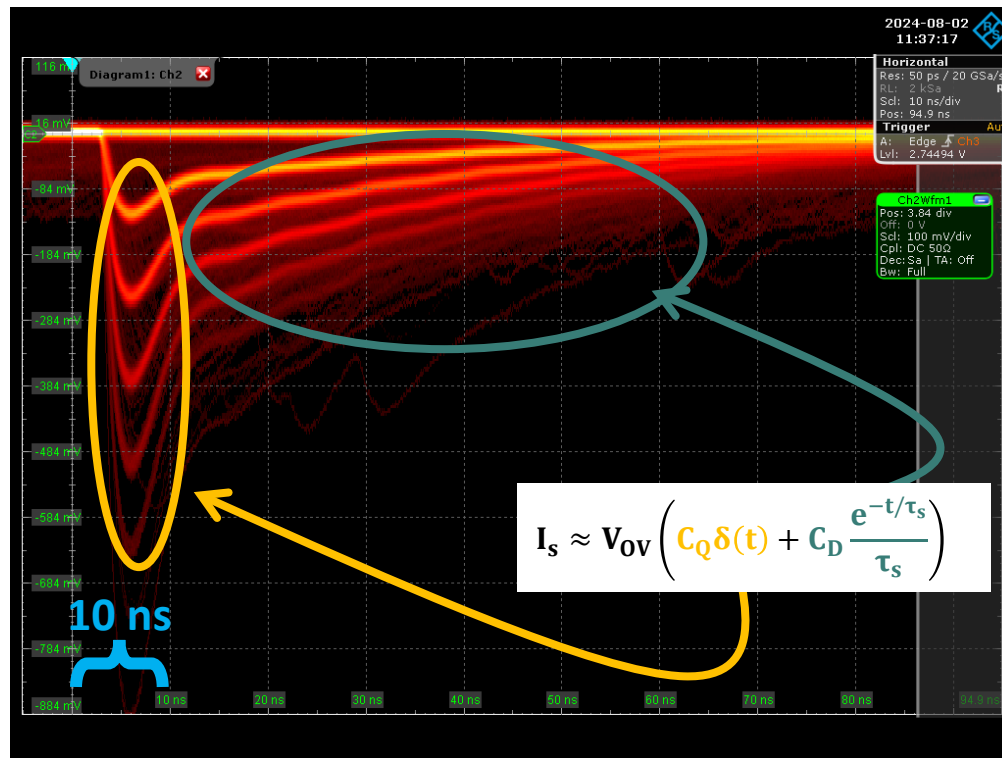


SiPM signal

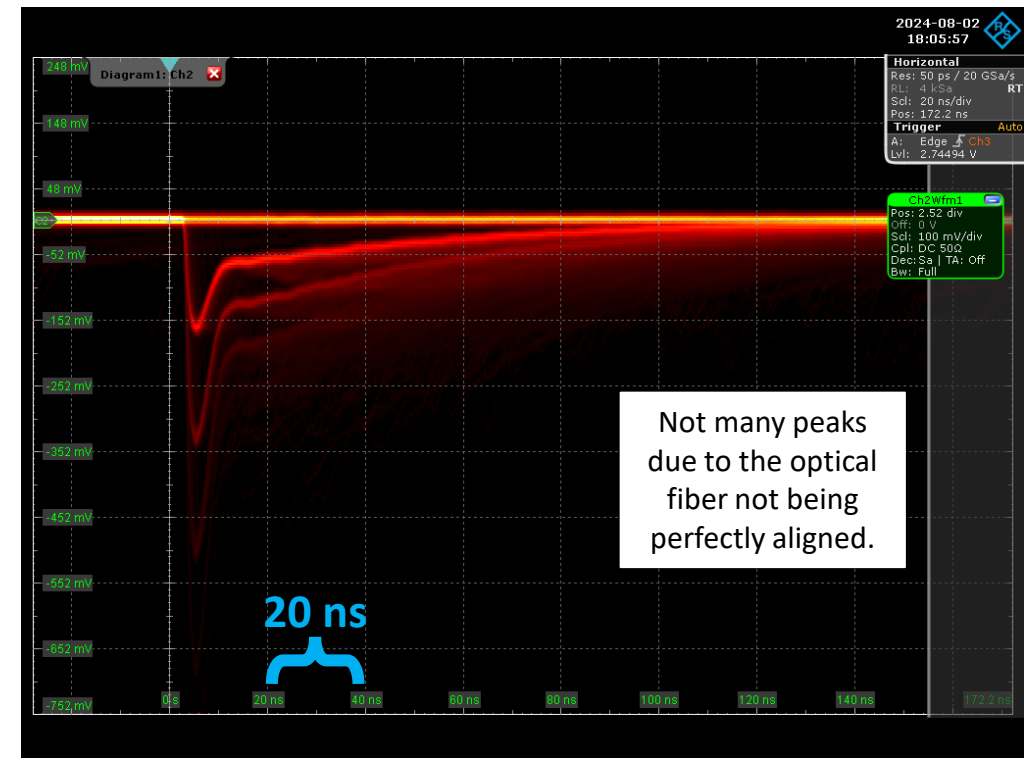
- The SiPM signal can be written as
$$I_s \approx V_{OV} \left(C_Q \delta(t) + C_D \frac{e^{-t/\tau_s}}{\tau_s} \right)$$
- The amplifier was tested with a 1.3 x 1.3 mm² Hamamatus SiPM (S13360-1350).
- Single photons were produced by a 405 nm laser controlled by the Hamamatsu PLP-10 pulser (70 ps FWHM pulses).
- Shown below: oscilloscope screenshots of photon counting with the SiPMs operated at 3 V_{OV}.



SiPM signal response (300 K, 3 V_{ov})

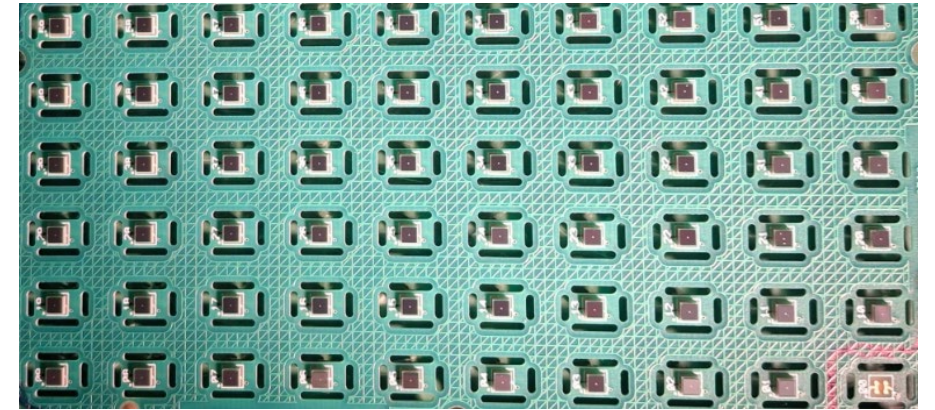
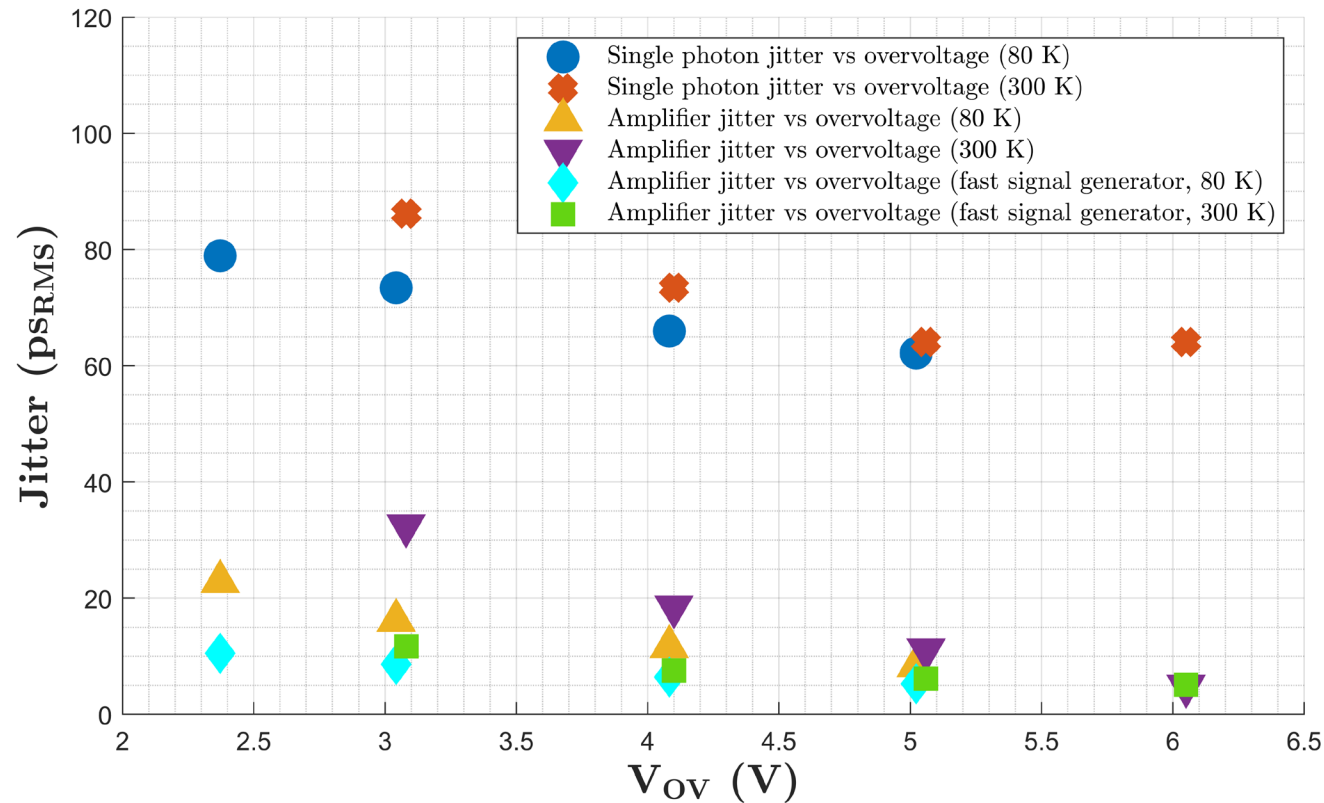


SiPM signal response (80 K, 3 V_{ov})



Single photon time resolution

- Measurement of the amplifier (amplifier jitter) and SiPM+amplifier (single photon time resolution)
- The effect of the amplifier is negligible compared with the SiPM at 300 K and 77 K, also at low overvoltage
- The amplifier is ready for the **joint R&D effort** to find the best SiPM candidate for LHCb RICH Upgrade 2
In collaboration with Ferrara, Genova, Padova, Perugia



Hamamatsu SiPMs mounted on carrier boards in Padova

These contributions have been removed:

- Signal generator
- Oscilloscope
- Light pulser

Thank you!